

II CALL 2022 (17/2/22)

DC)

$$I_{C2} = \frac{V_0}{R_C} = 1,88 \text{ mA}$$

$$I_{B2} = \frac{I_{C2}}{\beta_{F2}} = 13,5 \mu\text{A}$$

$$I_{E2} = I_{B2}(\beta_{F2} + 1) = 1,89 \text{ mA}$$

$$V_{BB} = V_{CC} \frac{R_{B2}}{R_{B1} + R_{B2}} = 14,26 \text{ V} ; R_{BB} = R_{B1} // R_{B2} = 48,7 \text{ K}\Omega$$

$$I_{B1} = \frac{V_{BB} - 0,7 - \beta_{F2} R_{E1}}{R_B + (\beta_{F1} + 1) R_{E1}} = 16,2 \mu\text{A}$$

$$I_{C1} = I_{B1} \beta_{F1} = 3,25 \text{ mA}$$

$$V_{E2} = V_{BB} - R_{BB} I_{B1} - V_{BE1} - V_{BE2} = 13,47 \text{ V}$$

$$R_{E2} = \frac{V_{CC} - V_{E2}}{I_{E2}} = 5,58 \text{ K}\Omega$$

$$V_{CE1} = V_{CC} - ((\beta_{F1} + 1) I_{B1} + I_{B2}) R_{E1} = 11,23 \text{ V}$$

$$V_{CE2} = V_0 - V_{E2} = -2,97 \text{ V}$$

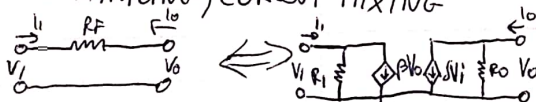
AC)

$V_T \approx 26 \text{ mV}$ THERMAL VOLTAGE

$$g_{m1} = \frac{I_{C1}}{26 \text{ mV}} = 125 \text{ mS} ; r_{\pi 1} = \frac{\beta_{01}}{g_{m1}} = 1,6 \text{ K}\Omega$$

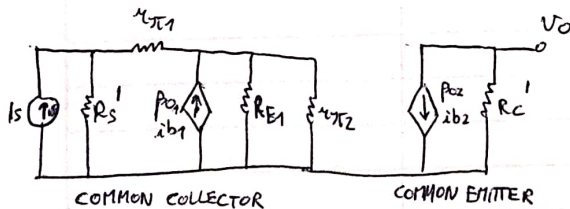
$$g_{m2} = \frac{I_{C2}}{26 \text{ mV}} = 72 \text{ mS} ; r_{\pi 2} = \frac{\beta_{02}}{g_{m2}} = 2,08 \text{ K}\Omega$$

VOLTAGE SAMPLING, CURRENT MIXING



$$R_1 = \frac{V_i}{i_1} \Big|_{V_o=0} = R_F \quad \beta = \frac{i_1}{i_o} \Big|_{V_i=0} = -\frac{1}{R_F}$$

$$R_o = \frac{V_o}{i_o} \Big|_{V_i=0} = R_F \quad \downarrow \rightarrow 0$$



$$I_S = \frac{V_S}{R_S}$$

$$R_S' = R_S // R_{BB} // R_F = 892,4 \Omega$$

$$R_C' = R_C // R_F = 3,59 \text{ K}\Omega$$

$$R_{L1} = r_{\pi 2} // R_{E1} = 1,36 \text{ K}\Omega$$

$$R_{in1} = r_{\pi 1} + (\beta_{01} + 1) R_{L1} = 2741,3 \text{ K}\Omega ; R_{out1} = R_{E1} // \left(\frac{R_S' + r_{\pi 1}}{\beta_{01} + 1} \right)$$

$$R_N = R_S' // R_{in1} = 889,5 \Omega$$

$$A_{u1} = R_N \cdot A_{vcc} = R_N \cdot \frac{(\beta_0 + 1) R_{L1}}{r_{\pi} + (\beta_0 + 1) R_{L1}} = 884,3 \Omega$$

$$R_{L2} = R_C' = 3,59 \text{ K}\Omega$$

$$A_{v2} = -g_{m2} R_{L2} = -258,9 \quad \text{BECAUSE } R_{E2} = 0$$

$$A_{oL} = A_{u1} \cdot A_{v2} = -228,9 \text{ K}\Omega$$

$$A_F = \frac{A_{oL}}{1 + \beta A_{oL}} = -9,58 \text{ K}\Omega$$

$$A_V = A_F \cdot \frac{1}{R_S} = -9,58$$

$$R_{in}^F = \frac{R_N}{1 + \beta A_{oL}} = 37,23 \Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}^F} + \frac{1}{R_S}} = 38,67 \Omega$$

$$R_{CS} = R_S + R_{BB} // R_F // R_{in1} = 9,05 \text{ K}\Omega$$

$$R_{CF1} = R_F + R_S // R_{BB} // R_{in1} = 10,98 \text{ K}\Omega$$

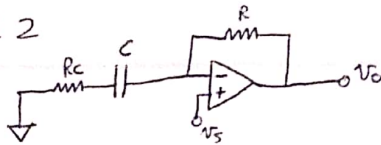
$$R_{CF2} = R_F + R_C = 15,6 \text{ K}\Omega$$

$$R_{CE} = \frac{1}{\frac{1}{R_{E2}} + \frac{\beta_0 + 1}{R_{OUT1} + r_{\pi 2}}} = R_{E2} // \left(\frac{R_{OUT1} + r_{\pi 2}}{\beta_0 + 1} \right) = 13,82 \Omega$$

$$f_L = \frac{1}{2\pi} \left(\frac{1}{R_{CS} C_S} + \frac{1}{C_F} \frac{1}{R_{CF1} + R_{CF2}} + \frac{1}{R_{CE} C_E} \right) = 11,51 \text{ KHz}$$

$$f_L^F = \frac{f_L}{1 + \beta A_{oL}} = 481,9 \text{ Hz}$$

EXERCISE 2



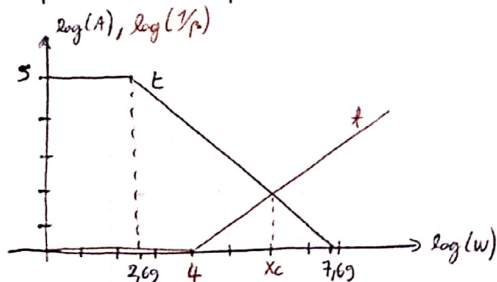
1) It's a non-inverting amplifier, so $A_{ideal}(s) = 1 + \frac{R}{Z_c} = 1 + RCs$ with $Z_c = \frac{1}{sC}$

$$2) \beta = \frac{v_-}{v_o} \Big|_{v_s=0} = \frac{Z_c}{Z_c + R} = \frac{1}{1 + RCs}$$

$$P\beta = \frac{1}{RC} = 10 \text{ K} \frac{\text{rad}}{\text{s}}$$

$$\beta(0) = 1 \rightarrow \frac{1}{P\beta} = 1$$

$$\beta(\infty) = 0 \rightarrow \frac{1}{P\beta} = \infty$$



$$t: y = -x + 7.69$$

$$f: y = x - 4$$

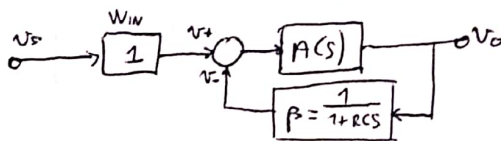
$$\Rightarrow x - 4 = -x + 7.69 \Leftrightarrow 2x = 11.69 \Leftrightarrow x = 5.85$$

$$\rightarrow X_c = 10^{5.85} = 708 \text{ K} \frac{\text{rad}}{\text{s}}$$

$$PM = 180 - \arctan\left(\frac{X_c}{w_0}\right) - \arctan\left(\frac{X_c}{P\beta}\right) = 0.85^\circ$$

3) It's a non-inverting amplifier, so it's a voltage sampling - voltage mixing

$W_{in} = 1$ in a non-inverting amplifier



$$4) Z_c = R_c + \frac{1}{sC} = \frac{R_c sC + 1}{sC}$$

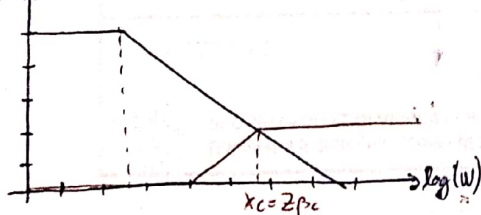
$$\beta_c = \frac{v_-}{v_o} \Big|_{v_s=0} = \frac{Z_c}{Z_c + R} = \frac{R_c sC + 1}{(R + R_c)sC + 1}$$

$$P\beta_c = \frac{1}{(R_c + R)C} \rightarrow \text{I NOTICE THAT IF } R_c \ll R \text{ THEN } P\beta_c \approx P\beta = \frac{1}{RC}$$

$$Z\beta_c = \frac{1}{R_c C} \quad \text{SO I KEEP THE POLE (ALMOST) IN THE SAME POSITION AND I PLACE A ZERO IN } w = X_c$$

$$I) \text{ impose } Z\beta_c = X_c \Leftrightarrow X_c = \frac{1}{R_c C} \Leftrightarrow R_c = \frac{1}{X_c C} = 141 \Omega$$

5) $\log(A), \log(1/P)$



$$P\beta_c = 9.86 \text{ K} \frac{\text{rad}}{\text{s}} \approx P\beta$$

$$PM = 180 - \arctan\left(\frac{X_c}{w_0}\right) - \arctan\left(\frac{X_c}{P\beta_c}\right) + \arctan\left(\frac{X_c}{Z\beta_c}\right) = 45.84^\circ$$

$$6) \text{ I PUT IN PARALLEL TO } R \text{ A CAPACITOR } C_c = \frac{1}{R X_c}, P'_c = \frac{1 + C_c R s}{(C + C_c) R s + 1}$$